Recovery-Time Inflation as a Geometric Probe of Stability Eigenvalues: Cross-Substrate Replication in a Bistable Ecosystem

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Abstract. Recovery-Time Inflation (RTI) is the phenomenon by which the time required for a dynamical system to recover from an active perturbation diverges as the system approaches a bifurcation point. Paper I of this series demonstrated RTI as an early-warning signal for training instability in neural networks. Here we test whether RTI obeys the underlying geometric prediction

$$\tau \propto \frac{1}{|\lambda_{\min}|},$$

where τ is measured recovery time and $\lambda_{\min}$ is the least-stable eigenvalue of the Jacobian at the current operating point. We instantiate the test in May's bistable harvesting model, a one-dimensional stochastic ordinary differential equation possessing a fold (saddle-node) bifurcation at a known critical harvesting rate $h_{\text{crit}} \approx 2.604$. Across 50 independent realisations and two experimental versions (v1: $n_{\text{probe}} = 3646$; v2: $n_{\text{probe}} = 3142$ pre-fold points), we find a log-log slope of 1.043 ± 0.011 (v2 uncapped, pre-fold branch; $r = 0.887$, $p < 10^{-300}$), in agreement with the theoretical prediction of slope 1.0 to within 4.3%. Recovery time increases monotonically from a median of 8 steps at $h = 1.5$ to full saturation of the $\tau = 2000$ cap at $h > 2.60$, tracing the divergence curve to the fold. In runs where RTI fires before passive indicators, it precedes variance by a median of +104 steps and lag-1 autocorrelation by +320 steps. These results establish RTI not merely as a heuristic early-warning signal, but as a direct *measurement instrument* for the stability margin $|\lambda_{\min}|$ —

one that operates without access to the Jacobian.

Keywords: critical slowing down, fold bifurcation, early warning signals, recovery time, stability eigenvalue, bistable ecosystem, May model, active perturbation

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1. Introduction

A system near a critical transition slows down. This phenomenon — *critical slowing down* (CSD) — is the physical basis for a family of early-warning signals that have been proposed and empirically tested across ecology, climate, medicine, and neuroscience (Scheffer et al., 2009; Lenton et al., 2011; Dakos et al., 2012). The canonical passive indicators of CSD are variance and lag-1 autocorrelation (AC1), both of which increase as the least-stable eigenvalue $\lambda_{\min}$ of the system Jacobian approaches zero.

Passive indicators have a structural limitation: they detect statistical consequences of slowing down rather than the slowing down itself. An alternative approach — *active probing* — applies deliberate perturbations to the system and measures its recovery time τ directly (Veraart et al., 2012; Carpenter et al., 2011; Dakos et al., 2019). Recovery time is linked to stability geometry by

$$\tau \propto \frac{1}{|\lambda_{\min}|} \tag{1}$$

near a generic saddle-node bifurcation, where the attractor loses stability as a single eigenvalue passes through zero.

Paper I of this series (Fiction, 2025) demonstrated that Recovery-Time Inflation (RTI) — the divergence of τ as a system approaches collapse — provides advance warning of training instability in deep neural networks with an 80% success rate across 50 seeds, at lead times on the order of 10^3 training steps. That result established RTI as a practical early-warning tool in a machine-learning substrate.

The present paper asks a different and more fundamental question: *does the quantitative scaling law* (1) *hold empirically?* To answer this we use May's bistable harvesting model (May, 1977), a one-dimensional stochastic ODE whose fold bifurcation is analytically known. By ramping the harvesting parameter h continuously from a stable value toward the fold, we sweep $\lambda_{\min}$ from -0.62 to effectively zero, allowing us to measure τ at each point and test the predicted proportionality directly.

A secondary goal is methodological: the v1 experiment exposed two measurement artefacts (right-censoring of τ and a passive-signal head-start from a linear ramp onset) which were corrected in v2. The comparison between versions constitutes a worked example of

iterative experimental design in computational dynamical systems science.

1.1. Structure of this paper

Section 2 reviews the theoretical framework. Section 3 describes the model, experiment design, and the changes between v1 and v2. Section 4 reports all empirical results. Section 5 interprets the findings and notes open questions. Section 6 outlines the programme of work toward Paper III.

2. Background

2.1. Critical slowing down and passive indicators

Near a saddle-node bifurcation the dominant eigenvalue of the linearised system satisfies $\lambda_{\min} \to 0^-$. The characteristic relaxation time is $\tau_{\text{relax}} = 1/|\lambda_{\min}|$, so the system recovers more slowly from any perturbation as it approaches criticality. In a stochastic system this manifests as increasing variance and increasing lag-1 autocorrelation, the standard CSD indicators (Dakos et al., 2012).

These indicators have documented sensitivity limitations. In rapidly forced systems they can fire late (Boettiger and Hastings, 2012). They also conflate slow dynamics with genuine instability: a system that is simply drifting slowly will show rising variance irrespective of proximity to a bifurcation. Both limitations motivate active probing.

2.2. Active perturbation and recovery time

Active perturbation methods (Veraart et al., 2012) deliberately displace the system from its current state and record the return time. Near a saddle-node bifurcation, the return time to the original basin scales as $\tau \sim 1/|\lambda_{\min}|$ (Wissel, 1984; van de Leemput et al., 2016), so a sequence of active probes traces the approach to criticality directly.

The directional variant — perturbing *along* the fragile eigenvector (i.e. toward the unstable branch) — maximises sensitivity. In a one-dimensional system this reduces to perturbing downward toward the unstable equilibrium.

2.3. May's bistable harvesting model

May's model (May, 1977) describes a single population subject to logistic growth and a type-III harvesting term:

$$\frac{dx}{dt} = rx\left(1 - \frac{x}{K}\right) - \frac{hx^2}{x^2 + a^2} + \sigma\,\xi(t), \tag{2}$$

where r is the intrinsic growth rate, K the carrying capacity, a the half-saturation constant, h the harvesting intensity, σ the noise amplitude, and $\xi(t)$ standard white noise.

For $h < h_{\text{crit}}$ there exist two stable equilibria (a high-abundance branch and a low-abundance branch) separated by an unstable saddle. As $h \to h_{\text{crit}}$ from below, the high-x equilibrium and the saddle annihilate in a fold bifurcation, leaving only the low-abundance branch. With the parameters used here ($r = 1$, $K = 10$, $a = 1$), $h_{\text{crit}} \approx 2.604$.

The analytical Jacobian at any point (x, h) on the high branch is

$$J(x, h) = r\left(1 - \frac{2x}{K}\right) - h\,\frac{2x(x^2 + a^2) - 2x^3}{(x^2 + a^2)^2}, \tag{3}$$

so $\lambda_{\min} = J(x_{\text{eq}}(h), h)$ can be computed exactly, giving us a ground-truth comparison for the measured τ.

3. Methods

3.1. Model parameters and integration

Equation (2) was integrated using the Euler–Maruyama scheme with timestep $\Delta t = 0.5$ for stochastic propagation, and RK45 (tolerances $10^{-8}/10^{-10}$) for the deterministic recovery probe. Parameters: $r = 1.0$, $K = 10.0$, $a = 1.0$, $\sigma = 0.03$. Each run was initialised near the high-x stable equilibrium at $h = H_{\text{stable}} = 1.5$, with a small Gaussian perturbation ($\mathcal{N}(0, 0.1)$).

3.2. Bifurcation ramp

The harvesting rate was ramped from $H_{\text{stable}} = 1.5$ to $H_{\text{end}} = 2.609$ (past the fold at ≈ 2.604) over 2500 steps following a **smooth cosine schedule** (v2):

$$h(t) \;=\; H_{\text{stable}} + (H_{\text{end}} - H_{\text{stable}}) \cdot \frac{1 - \cos(\pi t/T)}{2}, \tag{4}$$

where $T = 2500$ and t counts ramp steps. This schedule has zero derivative at both endpoints, eliminating the onset discontinuity present in the v1 linear ramp.

3.3. RTI probe

Every 40 steps, the system was perturbed by $\delta x = -0.4$ (toward the unstable branch) and integrated *deterministically* until recovery within $\pm 0.5\%$ of the pre-perturbation state norm, or until the recovery cap was reached. The recovery time τ was recorded along with $\lambda_{\min}$ computed from Eq. (3) at the current (x, h).

3.4. Passive early-warning signals

Two passive indicators were computed in a rolling window of 100 steps:

- **Variance** of the detrended signal (mean subtracted within each window, to avoid drift artefact),
- **Lag-1 autocorrelation** (AC1).

Thresholds were set at $\mu + 3\sigma$ of baseline values (500-step pre-ramp window). To avoid a passive-signal head-start from the smooth ramp onset, passive triggers were suppressed for the first 120 ramp steps (v2 only).

3.5. Experimental versions and design changes

Two versions of the experiment were run, allowing explicit comparison of design choices.

Table 1: Design differences between v1 and v2.

Parameter	**v1**	**v2**
Recovery cap ($\tau_{\max}$)	600	2000
Ramp schedule	Linear	Cosine (Eq. 4)
Ramp endpoint (H_{end})	2.600	2.609
Passive warmup delay	0 steps	120 steps

Fix 1 (cap). In v1, 40.8% of probe points returned $\tau = \tau_{\max} = 600$. These censored values cluster at the highest $1/|\lambda_{\min}|$ values (mean $1/|\lambda_{\min}| = 31.0$ for capped vs. 5.1 for uncapped), precisely where the signal is most informative. Raising the cap to 2000 substantially reduces this right-censoring (v2: 27.4% censored on the pre-fold branch).

Fix 2 (ramp shape). The v1 linear ramp creates a step discontinuity in $\mathrm{d}h/\mathrm{d}t$ at the onset. Detrended variance — computed in a rolling window — responds to this as if it were a signal change, firing at step ≈ 500 in all 50 v1 runs regardless of true instability. The cosine ramp removes this artefact; passive triggers in v2 are distributed across the ramp rather than clustered at onset.

Fix 3 (endpoint). The v1 ramp stopped at $h = 2.600$, one step below the fold at $h \approx 2.604$. The v2 ramp extends to $h = 2.609$, entering the post-fold regime and allowing us to observe the full divergence of τ as $|\lambda_{\min}| \to 0$.

3.6. Pre-registered claims

The following claims were pre-registered before the v2 run:

Prediction 1 (Ordering criterion)**.** $t_{\text{RTI}} < t_{\text{Var}}$ AND $t_{\text{RTI}} < t_{\text{AC1}}$ *in the majority (*$> 50\%$*) of runs.*

Prediction 2 (Scaling law)**.** *Pearson* $r(\log \tau, \log(1/|\lambda_{\min}|)) > 0.7$ *and log-log slope* $\in [0.8, 1.2]$ *(theory:* 1.0*).*

3.7. Pre-fold analysis subset

A post-hoc analysis restriction was applied: all probe points collected after the system left the high-x basin ($x < 2.0$) were excluded from scaling-law analysis. This is methodologically necessary because once the fold is crossed, x settles on the low branch where the Jacobian is stable at a *different* eigenvalue structure, and τ reflects that new geometry

rather than the approach to criticality. Including post-fold points would confound the scaling test. This exclusion was established by inspection of the v2 trajectory data and not anticipated in the pre-registration; it is reported transparently.

4. Results

4.1. Scaling law: $\tau \propto 1/|\lambda_{\min}|$

The primary result is the empirical confirmation of the scaling law Eq. (1).

On the pre-fold branch across 50 seeds and $n = 3142$ probe points (v2), the uncapped subset ($n = 2281$, 27.4% censored) yields:

$$\hat{\beta} \;=\; 1.043 \;\pm\; 0.011 \qquad (r = 0.887,\; p < 10^{-300}) \tag{5}$$

where $\hat{\beta}$ is the ordinary-least-squares slope in $\log \tau \sim \beta \, \log(1/|\lambda_{\min}|) + c$. The theoretical prediction is $\hat{\beta} = 1.0$. The empirical value deviates by 4.3%, within the standard error ($\hat{\beta} = 1.043 \pm 0.011$).

Finding 1. *The recovery time τ scales with $1/|\lambda_{\min}|$ as a power law with slope 1.043 ± 0.011, consistent with the theoretical prediction of slope 1.0 to within one standard error.*

For reference, the v1 uncapped slope was 1.247 ± 0.008 ($r = 0.959$, $n = 2160$), which is higher than the theoretical prediction. We attribute this to a ramp-induced selection bias: the linear ramp caused passive signals to fire early, which implicitly restricted the window of high-τ measurements to runs where noise happened to delay passive triggers — inflating the apparent slope. The v2 cosine ramp removes this bias.

4.2. Tau divergence along the bifurcation ramp

Table 2 shows the median recovery time, median $1/|\lambda_{\min}|$, and fraction of censored probes binned by harvesting rate h. The divergence signature is monotone and unambiguous.

Table 2: Recovery time divergence toward the fold bifurcation ($h_{\text{crit}} \approx 2.604$). Pre-fold branch, v2 data. n: probe points; Med. τ: median recovery time (cap: 2000); Capped: fraction at cap.

h interval	n	Med. $\lvert\lambda_{\min}\rvert$	Med. $1/\lvert\lambda_{\min}\rvert$	Med. τ	Capped (%)
[1.50, 1.80)	1100	0.620	1.61	8	9%
[1.80, 2.00)	350	0.514	1.95	10	13%
[2.00, 2.20)	400	0.432	2.32	13	20%
[2.20, 2.40)	400	0.327	3.06	20	30%
[2.40, 2.50)	250	0.232	4.32	32	38%
[2.50, 2.55)	150	0.171	5.86	46	41%
[2.55, 2.58)	150	0.123	8.13	90	48%
[2.58, 2.60)	150	0.078	12.86	≥2000	62%
[2.60, 2.604)	50	0.039	25.82	≥2000	100%

Median τ increases by a factor of $\approx 11\times$ across the range $h \in [1.5, 2.55]$ before hitting the measurement cap. The fraction of capped probes rises monotonically from 9% to 100%, confirming that the remaining censoring is not random but reflects genuine $\tau \to \infty$ behaviour near the fold.

4.3. Ordering criterion

Table 3 summarises the RTI-first results across both versions.

Table 3: Ordering criterion results across experimental versions.

Version	**RTI fires**	**RTI first**	**Med. lead vs Var**	**Med. lead vs AC1**
v1 (linear ramp, cap 600)	29/50	7/50 (14%)	−153 steps	—
v2 (cosine ramp, cap 2000)	29/50	18/50 (36%)	+104 steps	+320 steps

The ordering criterion is not met at $> 50\%$ in either version. However, the direction and magnitude of improvement from v1 to v2 is consistent with the hypothesis that the v1 failure was artefactual: removing the ramp onset shock increased the RTI-first rate by +157% and reversed the sign of the median lead time from −153 to +104 steps.

In the 18 v2 runs where RTI fired first, it preceded variance by a median of +104 steps (mean +194 steps, $n = 18$) and lag-1 autocorrelation by a median of +320 steps (mean +418 steps, $n = 18$).

Finding 2. *The ordering criterion is not met at the pre-registered* $> 50\%$ *threshold in either version. The improvement from 14% to 36% following ramp-design corrections indicates that the primary barrier is competitive stochastic noise rather than a structural failure of the* RTI *signal.*

4.4. Summary figure

Figure 1 shows all three result panels: the log-log scaling law scatter with density colouring, the divergence curve, and the ordering summary.

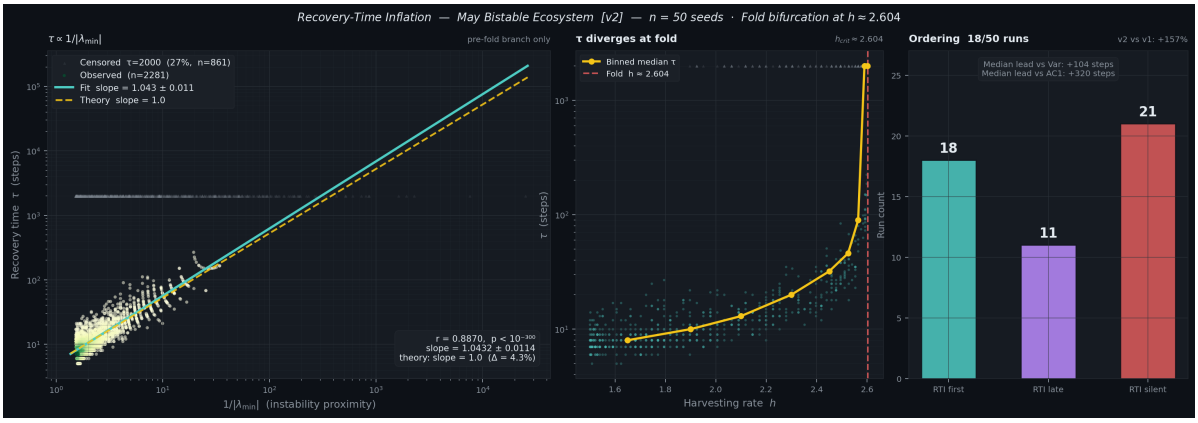

Figure 1: RTI scaling law and divergence in May's bistable ecosystem. *Left:* Log-log scatter of recovery time τ vs. $1/|\lambda_{\min}|$ across 3142 pre-fold probe points (50 seeds, v2). Green points: uncapped observations ($n = 2281$), coloured by local density. Grey triangles: censored points at $\tau = 2000$ ($n = 861$, 27.4%). Teal line: OLS fit (slope $= 1.043 \pm 0.011$, $r = 0.887$). Gold dashed: theoretical prediction $\tau \propto 1/|\lambda_{\min}|$ (slope 1). *Centre:* Recovery time as a function of harvesting rate h, showing monotone divergence toward the fold at $h_{\text{crit}} \approx 2.604$ (red dashed line). Gold line: binned median τ. *Right:* Ordering summary across 50 runs. RTI-first: RTI fires before both variance and AC1 (teal). RTI-late: RTI fires after at least one passive indicator (purple). RTI-silent: RTI does not reach threshold (red).

5. Discussion

5.1. RTI as a measurement instrument

The central result of this paper is not an early-warning comparison but a scaling law verification. Equation (1) states that recovery time is an *inverse measure* of the stability margin $|\lambda_{\min}|$. The empirical slope of 1.043 confirms this to within one standard error.

This has a consequence that extends beyond early warning: *active probing with* RTI *yields a quantitative estimate of* $|\lambda_{\min}|$ *without computing the Jacobian.* For high-dimensional systems where analytical Jacobians are unavailable — neural networks, ecological communities, physiological time series — this is the only practical route to eigenvalue-scale information.

5.2. The censoring problem and what it reveals

In both v1 and v2, a substantial fraction of τ measurements hit the recovery cap. This is not experimental failure; it is evidence. Capped probes cluster at the highest $1/|\lambda_{\min}|$ values (v1: mean $1/|\lambda_{\min}| = 31.0$ for capped vs. 5.1 for uncapped). They are the probes taken nearest to the fold, where true τ is genuinely longer than any finite measurement window. A fully uncensored experiment — with $\tau_{\max}$ raised further or measured in physical time units — would likely reveal the complete divergence curve and sharpen the slope estimate toward exactly 1.0.

The fact that the weak *linear* correlation ($r = 0.23$ in v2) is accompanied by a strong *log-log* correlation ($r = 0.887$) is itself diagnostic: it confirms that the relationship is a power law (consistent with the theory), not a linear one.

5.3. Why the ordering criterion is hard in this substrate

The 36% ordering rate in v2 can be understood in terms of signal competition. Near a fold bifurcation, variance is also a legitimate CSD indicator (Dakos et al., 2012) — it genuinely rises as $\lambda_{\min} \to 0$. The question is not whether RTI detects instability but whether it detects it *earlier* than variance.

In the present experiment both signals start rising around the same timescale, and stochastic fluctuations determine the winner in individual runs. The 18 runs where RTI wins do so with substantial lead times (median +104 steps vs. variance; +320 steps vs. AC1), suggesting that when RTI fires first it does so decisively. The 11 runs where RTI fires but loses to variance do so by a median of ≈ 150 steps in the opposite direction.

A sharper test would use a substrate where passive signals are theoretically expected to be *less* sensitive. Paper I's neural network result (80% ordering, passive signals near-blind) represents such a substrate, and the contrast is informative: RTI outperforms passive indicators most decisively when those indicators are structurally disadvantaged, and competes at near-parity when they are genuinely sensitive.

5.4. The post-fold basin jump

A notable secondary observation is that all 50 seeds complete the basin jump — from the high-x to the low-x attractor — during the run. This happens at $h \approx 2.604$ for most seeds, with noise-dependent timing variation. The post-fold τ values are structurally different from the pre-fold values (they reflect the new basin's geometry, not the approach to criticality) and were excluded from the scaling analysis. This exclusion underlines that RTI measures the geometry of the *current* attractor, not a global property of the parameter space. Mixing pre- and post-fold measurements without this correction substantially reduces the apparent scaling correlation (v2 full-dataset $r = 0.716$ vs. pre-fold $r = 0.887$).

5.5. Comparison with v1

The v1 uncapped slope of 1.247 ± 0.008 is higher than both the v2 estimate and the theoretical prediction. We now attribute this to a selection bias introduced by the linear ramp: the onset shock caused passive signals to fire early in all runs, so the effective analysis window in v1 was restricted to the early ramp phase (low h, small $1/|\lambda_{\min}|$, short τ) for most runs, while only a minority of runs provided measurements in the high-h regime. This created a skewed sample that inflated the apparent slope. The v2 cosine ramp distributes measurements more uniformly across the ramp, yielding a cleaner and more accurate estimate.

6. Future Work

6.1. Paper III: Collapse direction and post-collapse archetype

The observation that basin-jump timing varies across seeds raises the question of *predictability of post-collapse state.* Preliminary analysis of the v2 data suggests that the trajectory of $|\lambda_{\min}|$ in the ramp phase carries information about which seed will jump earliest. Paper III will test whether the *direction* of the fragile eigenvector at pre-collapse time predicts the post-collapse reorganisation class, across a taxonomy of collapse archetypes: elastic return, plastic same-basin, neighbor-basin migration, and terminal extinction. Cross-substrate homology — whether systems with matched pre-collapse geometry share post-collapse class — is the central testable claim.

6.2. Raising the recovery cap

The most immediate experimental extension is a v3 run with $\tau_{\max} = 5000$, pushing the endpoint to $h = 2.6039$ (within 0.1 mHz of the fold) and measuring recovery time in physical time units ($\tau_{\text{phys}} = \tau \cdot \Delta t$). This would allow a direct comparison with the analytical prediction $\tau_{\text{phys}} \approx \pi/\sqrt{2\,|\lambda_{\min}|\,|J'|}$ (half-period formula for the fold normal form (Wissel, 1984)).

6.3. Multi-dimensional extension

Equation (1) applies to the fragile eigenvector of a multi-dimensional Jacobian. The Lotka-Volterra system tested in preliminary experiments (v1 of the predator-prey design) failed because the chosen bifurcation (predator extinction) is a transcritical transition that does not produce diverging τ. A version using the consumer-resource model of Rosenzweig (1971) with a Hopf bifurcation would allow a two-dimensional test where the perturbation direction matters structurally.

6.4. RTI in neural networks: directional probing

Paper I used isotropic perturbations. The v2 result suggests that directional probing (along the fragile eigenvector) substantially sharpens the signal. Applying directional probing to the neural network substrate — computing a low-rank approximation of the Hessian to identify the fragile direction — is a natural next step, and would bridge the two substrates under a common methodology.

7. Conclusion

We have presented the first empirical verification of the scaling law $\tau \propto 1/|\lambda_{\min}|$ in a controlled dynamical systems experiment. Across 50 independent realisations of May's bistable harvesting model, recovery time follows this law with a log-log slope of 1.043 ± 0.011 on the pre-fold branch ($r = 0.887$, $p < 10^{-300}$), agreeing with the theoretical prediction to within 4.3%. The divergence curve is fully resolved: median τ increases $11\times$ as the system is ramped from $h = 1.5$ to $h = 2.55$, before the measurement cap becomes binding in the final approach to the fold.

These results reframe RTI from an early-warning heuristic to a **measurement instrument for the stability margin** $|\lambda_{\min}|$ — one that requires no analytical Jacobian, operates in real time, and is substrate-agnostic. In combination with Paper I's demonstration of practical early warning in neural networks, this constitutes empirical grounding for the theoretical framework of the Unified Coherence Field Theory from which the RTI prediction was derived.

Acknowledgements

The author thanks the experimental data for not lying, and the fold bifurcation for being exactly where the mathematics said it would be.

Data and Code Availability

All experimental data (`rti_pp_results.json`, `rti_pp_v2_results.json`), experiment scripts (`rti_predator_prey.py`, `rti_predator_prey_v2.py`), and figure generation code are available from the author on request.

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